

# Identifying Possible Pathogenic Microbial Species in Ulcerative Colitis

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## Abstract

Metagenomics is a common approach to the study of inflammatory bowel disease (IBD), since dysbiosis is a well-known co-occurrence. Whether pathogenesis is of microbial origin or due to the patient's lifestyle, genetics, and environment is unclear. For my research project, I investigated microbiome constituents specifically in ulcerative colitis (UC) to discover what species are hypothetically pathogenic. After comparing highly abundant microbial species in UC against a control group, I found the following four species to be five times more highly represented in UC: *Clostridium nexile*, *Parabacteroides* (unclassified), *Bifidobacterium animalis*, and *Ruminococcus gnavus*.

## Background

Inflammatory bowel disease (IBD) is a multifactorial chronic condition of the gastrointestinal (GI) tract, characterized by flares of inflammation. The condition has been attributed to a deregulated immune response to gut microbiome dysbiosis, although the exact cause of the condition is unknown. Across sub-types of IBD and established in studies, there is an observed depletion in microbiome diversity.

IBD has been differentiated into sub-types depending on what part of the GI tract becomes inflamed and the physiology of the diseased tissue. Crohn's disease mainly affects the small intestine and is characterized by chronic inflammation, as indicated by consistent high levels of fecal-calprotectin concentration ([f-c]) in the samples. Ulcerative colitis (UC) mainly affects the colon and is characterized by flares of inflammation, so the [f-c] varies across samples within this diagnosis (Halfvarson).

Researchers at Harvard and Massachusetts General Hospital (MGH) performed a very broad multi'omic study on a cohort from the Prospective Registry in IBD Study at MGH (PRISM), which analyzed not only the microbial metagenomes from samples, but also determined differentially abundant enzymes and metabolites of both microbial and host origin (Franzosa). They did not take current levels of inflammation into account during their analysis, which is a significant factor in the study of IBD, especially UC. To expand on their work, I will group the samples by [f-c] and take a close look at the relative abundance (RA) of microbial species in UC.

## Results

To identify possible pathogens in UC, I analyzed the PRISM cohort relative abundance (RA) data for microbial species, acquired from the 2019 multi'omic study (Franzosa). In the original publication, [f-c] was not taken into consideration, but Halfvarson, et al. found that this was an important component to the UC diagnosis. I wrote a Python script (see Appendix 1) to group the microbial species RA based on diagnosis (UC, CD, or Control) and then separate each group based on [f-c], where samples with [f-c] > 100 µg/g were sorted into the high groups and [f-c] < 100 µg/g into low groups, resulting in a total of six subgroups. Interestingly, there were six samples in the high [f-c] control group, with concentrations ranging from 101 – 360 µg/g.

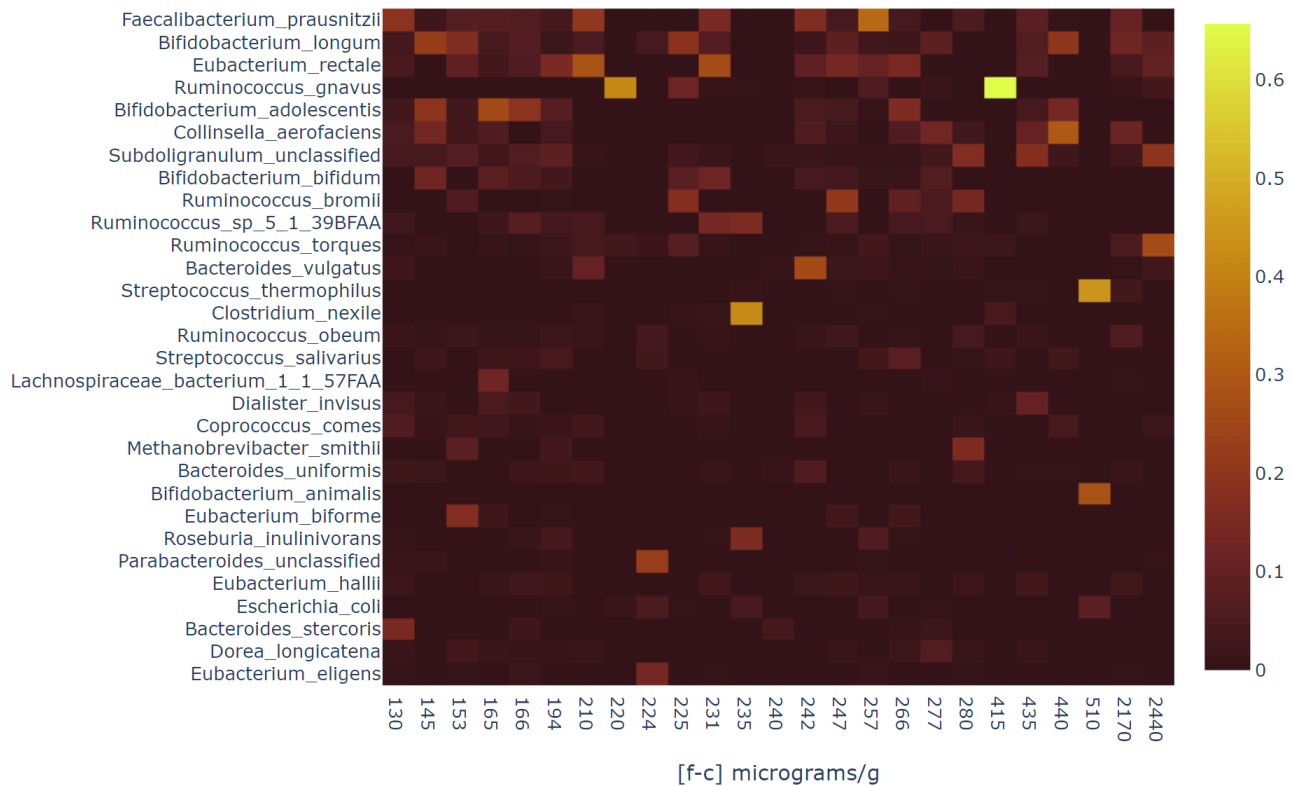
Statistics were for [f-c] were calculated across the entire sample set, for all low [f-c] and high [f-c] samples, for each diagnosis, and for both [f-c] subgroups within each diagnosis. The script prints all statistics neatly into one .txt file (see Appendix 2). There were 67 samples in the PRISM cohort that did not have [f-c] data. This is most likely because 101 investigators comprise the program that populates the biorepository the cohort was selected from, and there may be some inconsistency in how samples are processed (CSIBD, MGH).

To summarize the UC statistics, which is the primary target of my research: the range of the high [f-c] UC subgroup, n = 26, is 130 – 2440 µg/g, with a mean of 410 µg/g, median 238 µg/g, and std dev 568 µg/g. The range of the low [f-c] UC subgroup, n = 20, was 7 – 68 µg/g, with a mean of 35 µg/g, median 40 µg/g, and std dev 18 µg/g. For all 46 UC samples with [f-c] data, the mean was 247 µg/g, median 149 µg/g, and std dev 463 µg/g.

To determine what species are most prevalent in UC patients, I summed the RAs across each row and ranked them. The Python script saves these modified tables to .csv files and creates a .txt file containing a list of the top 30 RA species in each of the two UC disease state groups (Appendix 3). Those species that are more abundant in UC than in the controls are possible pathogens. I created a visual representation of the results for high [f-c] UC in Figure 1, where the UC samples are sorted in ascending order by [f-c] on the x-axis (left to right) and in descending order by total relative abundance on the y-axis (top to bottom). Since I used Plotly to generate the figures, when the script is run, they open in your web browser and are interactive (a GitHub link is provided in Appendix 1 if you would like to download the script).

The top three ranked species (*Faecalibacterium prausnitzii*, *Bifidobacterium longum*, and *Eubacterium rectale*) are abundant most samples, which may indicate that these are normal constituents in the human microbiome, regardless of disease state. However, there appears to be

an acute infection of other species that inflates the total RA. Notice the aberrant high RAs of *Ruminococcus gnavus*, *Streptococcus thermophilus*, *Clostridium nexile*, and *Bifidobacterium animalis*.



*Figure 1: The relative abundance across 26 samples for the top 30 species represented in the PRISM cohort subgroup diagnosed as UC with [f-c] > 100 µg/g; the samples are ordered by [f-c] on the x-axis, and species are ranked in descending order by total RA.*

To clearly differentiate between species which are normally highly abundant in the human microbiome from those uniquely enriched in UC, I calculated the average RA for each microbial species in the UC groups (n=26, n=20) and the low [f-c] control group (n=36). When the top 30 RA microbial species for low [f-c] and high [f-c] UC subgroups were combined, there were 38 unique species across the two lists. The script retrieves the average RA value for each of these 38 species from the high [f-c] UC and low [f-c] control groups and compares them. It then prints a list of the 18 species which were found to be more abundant in UC than in controls (Appendix 4), and these were graphed on a bar plot (Figure 2).

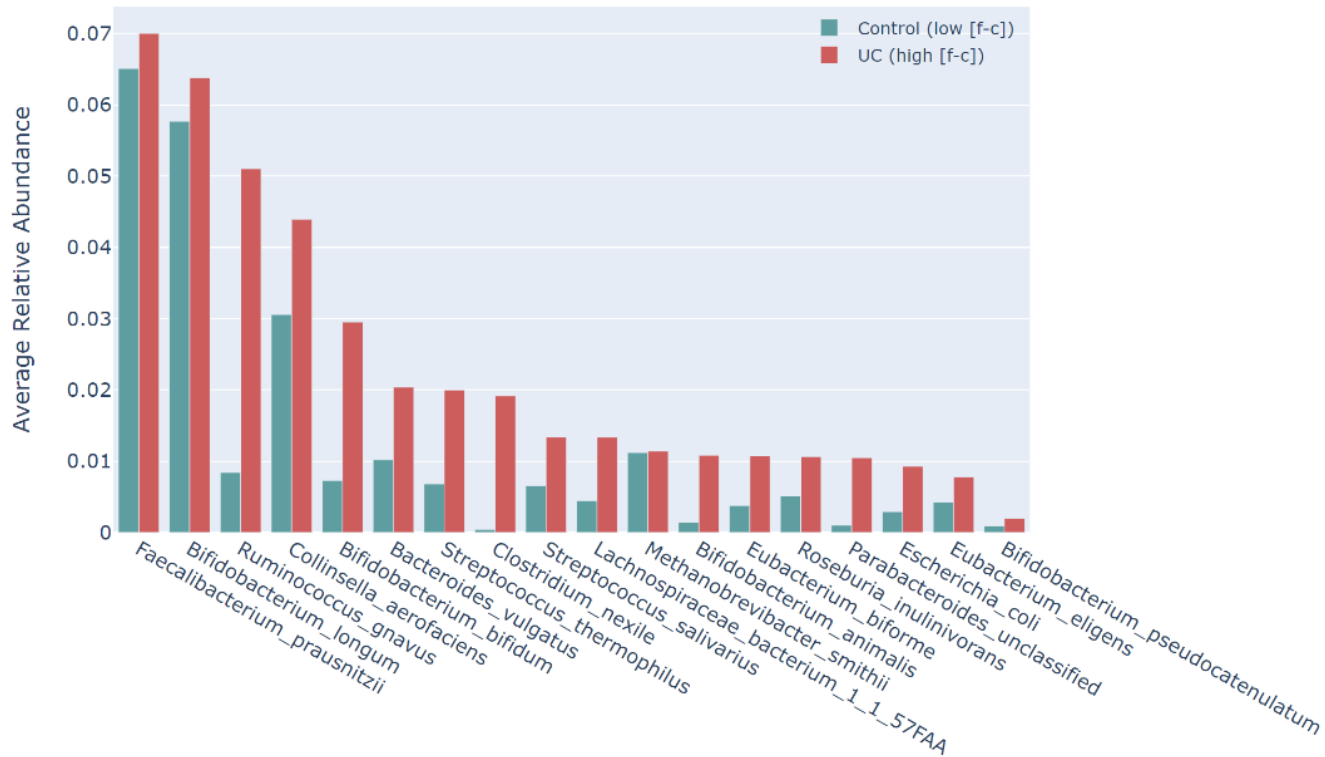


Figure 2: The average RA of the top 30 candidate species, discovered in the initial ranking by total RA, were compared against the PRISM control subgroup with  $[f-c] < 100 \mu\text{g/g}$ . The 18 species enriched in UC were graphed against corresponding control values. Since low biodiversity is a well-known trait in IBD, generally DA species are more likely to be depleted in UC, not enriched as seen here.

Plotting the average RA of microbial species in UC against the RA of those same species in the Control group confirmed that some of these are also significantly represented in healthy individuals. Possible explanations for the depletion in controls are that these beneficial microbes that are out-competed by UC pathogens, or they simply do not thrive in an inflamed-state gut environment. To further narrow in on species that are uniquely enriched in UC samples, the ratio of UC to Control RAs was calculated (Appendix 5, Figure 3).

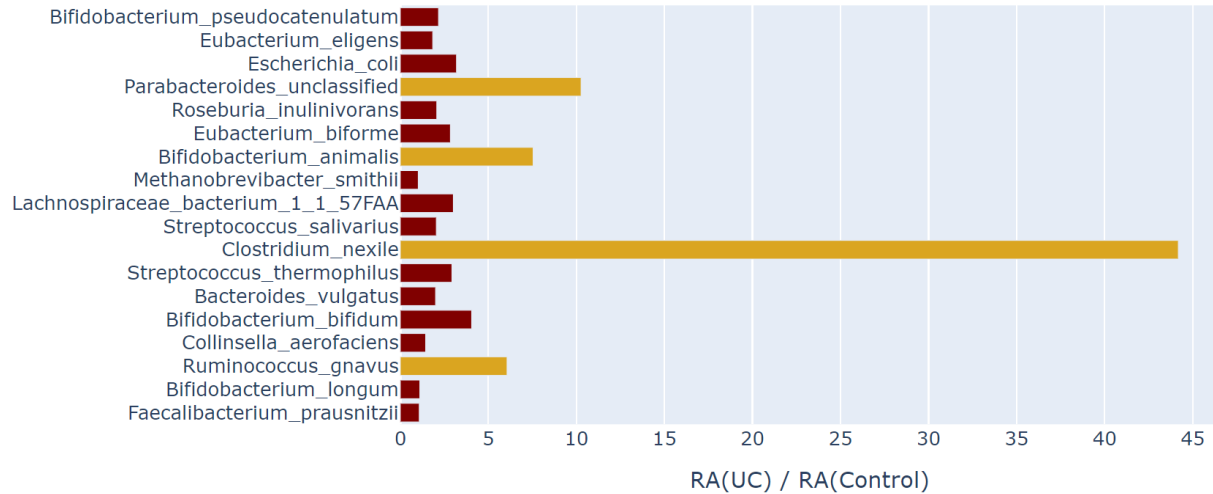


Figure 3: The ratio of RA in UC to RA in the control group was calculated and graphed. Species retain the same order as in Figure 2. The ratios > 5 are highlighted in yellow.

## Conclusion

Subsequent calculations, analysis and graphing narrowed the pool of 209 total microbial species present in the PRISM cohort down to just four putative pathogens found in UC patients: *Clostridium nexile*, *Parabacteroides* (unclassified), *Bifidobacterium animalis*, and *Ruminococcus gnavus*. These were not the DA species in UC indicated by Franzosa, et al, and of these four only *R. gnavus* was indicated in their paper as DA in CD (see Supplementary Figure).

The next step in this line of research would be to find if these are uniquely DA in UC or if they are shared by CD patients, and then suggest how each species may contribute to pathogenesis by reviewing the literature, conducting cell culture experiments, and associating them with DA metabolites. This research would also benefit from a more robust metagenomic dataset, which could be achieved by merging the PRISM cohort subgroup with other UC metagenomic datasets where [f-c] was measured (e.g. the Halfvarson data).

## Methods

All data processing, calculations, and graphing were performed using Python. The libraries I invoked were Pandas, Numpy, Plotly, Operator, and Statistics. The analysis was conducted on the PRISM cohort microbial RA data, which was provided by Franzosa, et al. as supplementary material to their publication.

## Works Cited

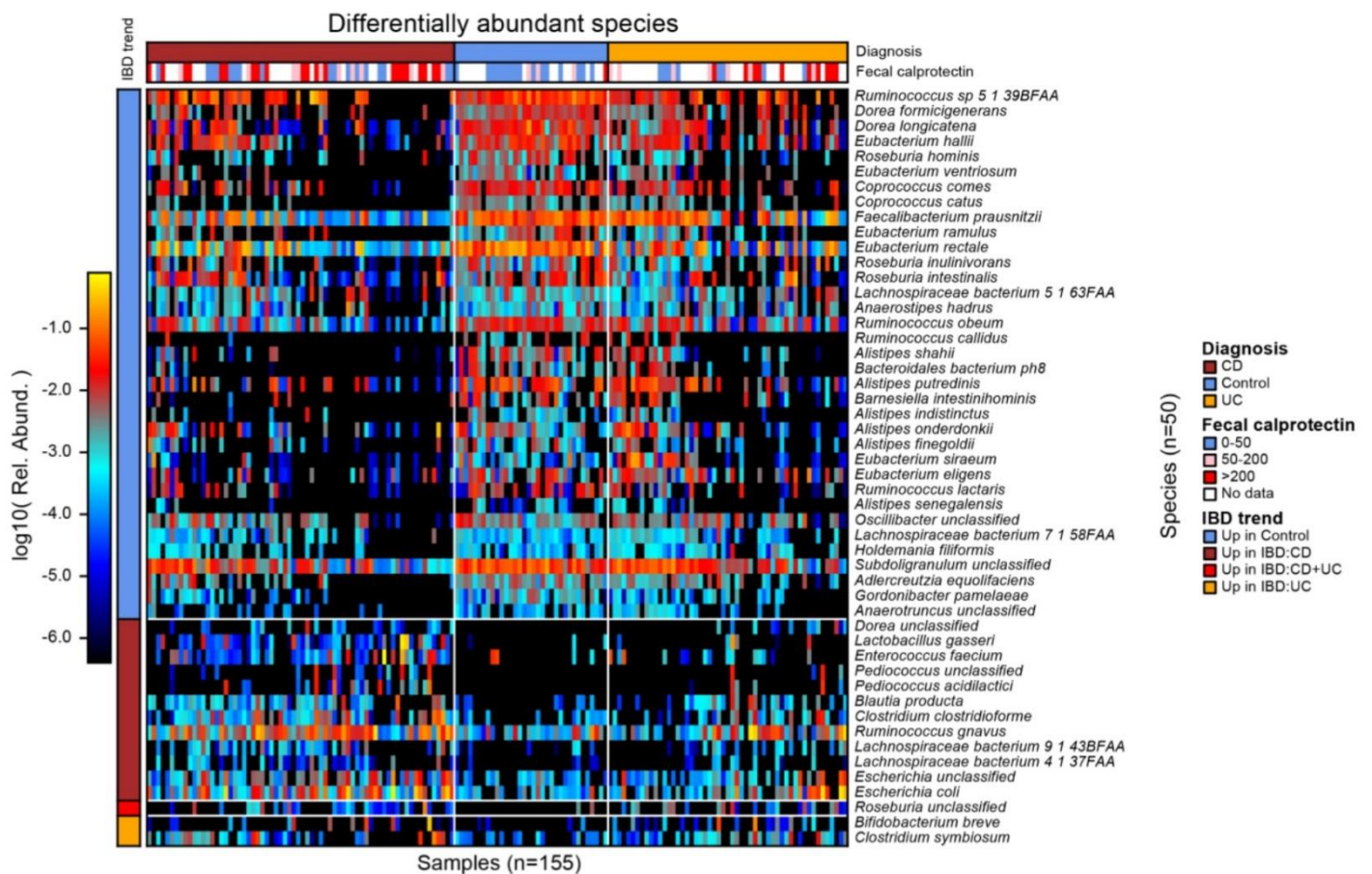
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## Resources

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## Supplementary Figure

Differentially abundant species in IBD. Fransoza, et al. named 12 DA species unique to CD (red), two unique to UC (yellow), and one unique to IBD (both CD and UC). Their conclusions differ from my own, which considered fecal-calprotectin concentration as a differentiating factor between samples. For UC, they found *Bifidobacterium breve* and *Clostridium symbiosum* to be DA. I found *Clostridium nexile* to be the most substantially DA species compared to the control group, while *Parabacteroides* (unclassified) and *Bifidobacterium animalis* were also significant. *Ruminococcus gnavus* and *Escherichia coli*, which I found to be highly abundant in UC, was indicated by Franzosa, et al. as also prominent in CD.



## Appendix 1: Python code

The script and dependencies are uploaded on [GitHub](#) if you would like to run it. When you do, the summaries will print to text files, and the interactive graphics will open in your web browser. See 'README.md' for more information.

```
# -*- coding: utf-8 -*-
"""
Created on Wed Apr 22 15:19:19 2020
@author: jlbrowne

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"""

import pandas as pd
import numpy as np
import operator as op
from plotly.offline import plot
import plotly.graph_objects as go
from statistics import mean, median, mode, stdev

##### DATA PROCESSING #####
def flt(value):
    _fc=[]
    for h in value:
        _fc.append(float(h))
    return _fc

def get_ft(sample):
    _ft = []
    for value in sample:
        _ft.append(op.itemgetter(0)(value))
    return _ft

# import abundance data for PRISM cohort
microb = pd.read_csv('microb_abund.csv', header=1, index_col=0)

# separate datasets based on [f-calprotectin] (fc) level
# and diagnosis (dg): Crohn's (CD), ulcerative colitis (UC)
fc = microb.loc['Fecal.Calprotectin']
fc = flt(fc)
dg = microb.loc['Diagnosis']
mg = microb.loc['SRA_metagenome_name']
ft = list(microb.columns)
sample_list = list(zip(ft, mg, dg, fc))

uc_hifc = []
uc_lofc = []
cd_hifc = []
cd_lofc = []
control_hifc = []
control_lofc = []
nofc = []
```



```

for i,j,k,l in sample_list:
    if k == 'UC':
        if l > 99.9:
            uc_hifc.append([i,j,l])
        if l < 100:
            uc_lofc.append([i,j,l])
    if k == 'CD':
        if l > 99.9:
            cd_hifc.append([i,j,l])
        if l < 100:
            cd_lofc.append([i,j,l])
    if k == 'Control':
        if l > 99.9:
            control_hifc.append([i,j,l])
        if l < 100:
            control_lofc.append([i,j,l])
    if np.isnan(l):
        nofc.append([i,j,k])

#verify all 220 samples are accounted for
divided = [uc_hifc, uc_lofc, cd_hifc, cd_lofc, control_hifc, control_lofc,
nofc]
counts = []
for each in divided:
    counts.append(len(each))
sum(counts)

# statistics for [f-c] in UC subgroups
hi_uc_fc = []
for value in uc_hifc:
    hi_uc_fc.append(op.itemgetter(2)(value))
lo_uc_fc = []
for value in uc_lofc:
    lo_uc_fc.append(op.itemgetter(2)(value))
hi_cd_fc = []
for value in cd_hifc:
    hi_cd_fc.append(op.itemgetter(2)(value))
lo_cd_fc = []
for value in cd_lofc:
    lo_cd_fc.append(op.itemgetter(2)(value))
lo_control_fc = []
for value in control_lofc:
    lo_control_fc.append(op.itemgetter(2)(value))
hi_control_fc = []
for value in control_hifc:
    hi_control_fc.append(op.itemgetter(2)(value))
all_control_fc = lo_control_fc + hi_control_fc
all_uc_fc = hi_uc_fc + lo_uc_fc
all_cd = hi_cd_fc + lo_cd_fc
all_hi = hi_uc_fc + hi_cd_fc + hi_control_fc
all_lo = lo_uc_fc + lo_cd_fc + lo_control_fc
every = all_hi + all_lo

def stats(fc):
    f.write('\nmean: ' + str(round(mean(fc),2)) + '\n')
    f.write('median: ' + str(median(fc)) + '\n')
    f.write('std dev: ' + str(round(stdev(fc),2)) + '\n')
    f.write('min: ' + str(min(fc)) + '\n')
    f.write('max: ' + str(max(fc)) + '\n')

```

```

        f.write('n=' + str(len(fc)) + '\n\n')

f = open('fc_statistics.txt', 'w')
f.write('Statistics for f-calprotectin concentration in each group: \n\n')
f.close()

f = open('fc_statistics.txt', 'a')
f.write('High [f-c] CD: ')
stats(hi_cd_fc)
f.write('Low [f-c] CD: ')
stats(lo_cd_fc)
f.write('High [f-c] UC: ')
stats(hi_uc_fc)
f.write('Low [f-c] UC: ')
stats(lo_uc_fc)
f.write('High [f-c] Control: ')
stats(hi_control_fc)
f.write('Low [f-c] Control: ')
stats(lo_control_fc)
f.write('All CD: ')
stats(all_cd)
f.write('All UC: ')
stats(all_uc_fc)
f.write('All Control: ')
stats(all_control_fc)
f.write('All High: ')
stats(all_hi)
f.write('All Low: ')
stats(all_lo)
f.write('All: ')
stats(every)
f.close()

##### ANALYSIS #####
# Microbioal species abundance in UC and Control
#create a list of # Feature / Sample for subgroups
uc_ft_hi = get_ft(uc_hifc)
uc_ft_lo = get_ft(uc_lofc)

#slice original tables using feature list
uc_hi_microb = microb[uc_ft_hi]
uc_lo_microb = microb[uc_ft_lo]

# Heatmap for UC grouped by [fc]
conc_fc_hi_uc = hi_uc_fc
conc_ft_hiuc = list(zip(uc_ft_hi,conc_fc_hi_uc))
conc_ft_hiuc = sorted(conc_ft_hiuc, key=op.itemgetter(1), reverse=True)
ft_hiuc = get_ft(conc_ft_hiuc)
hm_hiuc = uc_hi_microb[ft_hiuc]
hm_hiuc = hm_hiuc[8:209].astype(float)
hm_hiuc['total_RA'] = hm_hiuc.sum(axis=1)
hm_hiuc = hm_hiuc.sort_values('total_RA', ascending=False)
hm_z_hiuc = hm_hiuc.iloc[0:30,:26]
hm_z = hm_z_hiuc.to_numpy()
hm_y = hm_z_hiuc.index.to_list()
hm_x = sorted(conc_fc_hi_uc )

fig1 = go.Figure()

```

```

fig1 = go.Figure(
    data=go.Heatmap(
        z=hm_z,
        x=hm_x,
        y=hm_y,
        transpose=False,
        colorscale='Solar'),
    layout=go.Layout(
        xaxis=dict(type='category'),
        yaxis=dict(autorange='reversed'),
        xaxis_title="[f-c] micrograms/g"
    )
)

plot(fig1, show_link = True, filename = 'heatmap.html')

# narrow in on microbes uniquely enriched in UC
# sum RA across samples, rank, and divide by n
uc_hi_microb = uc_hi_microb[8:209].astype(float)
uc_hi_microb['total_RA'] = uc_hi_microb.sum(axis = 1)
uc_hi_microb = uc_hi_microb.sort_values('total_RA', ascending=False)
uc_hi_microb['avg_RA'] = uc_hi_microb['total_RA']/26

uc_lo_microb = uc_lo_microb[8:209].astype(float)
uc_lo_microb['total_RA'] = uc_lo_microb.sum(axis = 1)
uc_lo_microb = uc_lo_microb.sort_values('total_RA', ascending=False)
uc_lo_microb['avg_RA'] = uc_lo_microb['total_RA']/20

control_ft_hi = get_ft(control_hifc)
control_ft_lo = get_ft(control_lofc)
control_hi_microb = microb[control_ft_hi]
control_lo_microb = microb[control_ft_lo]

control_hi_microb = control_hi_microb[8:209].astype(float)
control_hi_microb['total_RA'] = control_hi_microb.sum(axis = 1)
control_hi_microb = control_hi_microb.sort_values('total_RA',
ascending=False)
control_hi_microb['avg_RA'] = control_hi_microb['total_RA']/6

control_lo_microb = control_lo_microb[8:209].astype(float)
control_lo_microb['total_RA'] = control_lo_microb.sum(axis = 1)
control_lo_microb = control_lo_microb.sort_values('total_RA',
ascending=False)
control_lo_microb['avg_RA'] = control_lo_microb['total_RA']/36

#print to csv files for future analysis
uc_hi_microb.to_csv('PRISM_uc_hifc_microb_abund.csv')
uc_lo_microb.to_csv('PRISM_uc_lofc_microb_abund.csv')
control_hi_microb.to_csv('PRISM_control_hifc_microb_abund.csv')
control_lo_microb.to_csv('PRISM_control_lofc_microb_abund.csv')

# print the top RA microbes to file
hi_uc_30 = uc_hi_microb.iloc[0:30,[27]]
lo_uc_30 = uc_lo_microb.iloc[0:30,[21]]
hi_uc_30.to_csv('PRISM_top_30_RA_microbes_inflamed_uc.txt', sep='\t')
lo_uc_30.to_csv('PRISM_top_30_RA_microbes_lofc_uc.txt', sep='\t')

# find microbes that are more abundant in UC than Control
def get_RA(ft, zipped):

```

```

RA = []
for i in ft:
    for x,y in zipped:
        if i == x:
            RA.append(y)
return RA

mc1 = hi_uc_30.index.to_list()
mc2 = lo_uc_30.index.to_list()
xuc = list(dict.fromkeys(mc1+mc2))

mc = uc_hi_microb.index.to_list()
ra = list(uc_hi_microb['avg_RA'])
mc_ra = list(zip(mc,ra))
y_hiuc = get_RA(xuc, mc_ra)

mc = control_lo_microb.index.to_list()
ra = list(control_lo_microb['avg_RA'])
mc_ra = list(zip(mc,ra))
y_locontrol = get_RA(xuc, mc_ra)

control_vs_uc = list(zip(xuc,y_locontrol,y_hiuc))
suspects = []
for i,j,k in control_vs_uc:
    if j < k:
        suspects.append([i,k,j])

f = open('PRISM_UC_suspects.txt', 'w')
f.write('These 18 species were found to be more abundant in UC than in
controls \n\n')
    'Species' + '\t' + 'UC (high [f-c])' + '\t' + 'Control (low [f-c])' +
'\n')
f.close()
f = open('PRISM_UC_suspects.txt', 'a')
for i,j,k in suspects:
    f.write(i + '\t' + str(round(j,5)) + '\t' + str(round(k,5)) + '\n')
f.close()

x_suspect = get_ft(suspects)
y_suspect_uc = []
y_suspect_control = []
for value in suspects:
    y_suspect_uc.append(op.itemgetter(1)(value))
    y_suspect_control.append(op.itemgetter(2)(value))

fig2 = go.Figure()
fig2.add_trace(go.Bar(
    x=x_suspect,
    y=y_suspect_control,
    name='Control (low [f-c])',
    marker_color='cadetblue'
))
fig2.add_trace(go.Bar(
    x=x_suspect,
    y=y_suspect_uc,
    name='UC (high [f-c])',
    marker_color='indianred'
))
fig2.update_layout(

```

```

    xaxis_tickfont_size=14,
    yaxis=dict(
        title='Average Relative Abundance',
        titlefont_size=16,
        tickfont_size=14,
    ),
    legend=dict(
        x=0.75,
        y=1.0,
        bgcolor='rgba(255, 255, 255, 0)',
        bordercolor='rgba(255, 255, 255, 0)'
    ))
plot(fig2, show_link = True, filename = 'avg_RA_plot.html')

ratios = np.asarray(y_suspect_uc) / np.asarray(y_suspect_control)
species_ratios = list(zip(x_suspect, ratios.tolist()))
f = open('PRISM_UC_vs_control_ratios.txt', 'w')
f.write('Ratio of high [f-c] UC species RA by low [f-c] control \n\n'
        'Species' + '\t' + 'Ratio' + '\n')
f.close()
f = open('PRISM_UC_vs_control_ratios.txt', 'a')
for i,j in species_ratios:
    f.write(i + '\t' + str(round(j,5)) + '\n')
f.close()

colors = ['maroon',] * 18
colors[2] = 'goldenrod'
colors[7] = 'goldenrod'
colors[11] = 'goldenrod'
colors[14] = 'goldenrod'

fig3 = go.Figure(go.Bar(
    x=ratios,
    y=x_suspect,
    orientation='h',
    marker_color=colors
))
fig3.update_layout(
    xaxis=dict(
        title='RA(UC) / RA(Control)',
        tick0=0,
        dtick=5)
)

plot(fig3, show_link = True, filename = 'ratio_plot.html')

```

## Appendix 2: Print out for [f-c] statistic calculations

Statistics for f-calprotectin concentration in each group:

High [f-c] CD:  
mean: 223.33  
median: 215.0  
std dev: 86.31  
min: 100.0  
max: 625.0  
n=36

All CD:  
mean: 139.52  
median: 145.0  
std dev: 114.71  
min: 2.0  
max: 625.0  
n=65

Low [f-c] CD:  
mean: 35.48  
median: 40.0  
std dev: 22.91  
min: 2.0  
max: 86.0  
n=29

All UC:  
mean: 246.7  
median: 149.0  
std dev: 463.09  
min: 7.0  
max: 2440.0  
n=46

High [f-c] UC:  
mean: 409.5  
median: 237.5  
std dev: 567.78  
min: 130.0  
max: 2440.0  
n=26

All Control:  
mean: 45.86  
median: 17.5  
std dev: 67.94  
min: 1.0  
max: 360.0  
n=42

Low [f-c] UC:  
mean: 35.05  
median: 40.0  
std dev: 18.01  
min: 7.0  
max: 68.0  
n=20

All High:  
mean: 290.24  
median: 221.5  
std dev: 366.08  
min: 100.0  
max: 2440.0  
n=68

High [f-c] Control:  
mean: 174.83  
median: 138.0  
std dev: 98.29  
min: 101.0  
max: 360.0  
n=6

All Low:  
mean: 30.67  
median: 24.0  
std dev: 23.87  
min: 1.0  
max: 98.0  
n=85

Low [f-c] Control:  
mean: 24.36  
median: 14.5  
std dev: 26.45  
min: 1.0  
max: 98.0  
n=36

All:  
mean: 146.03  
median: 64.0  
std dev: 275.92  
min: 1.0  
max: 2440.0  
n=153

### Appendix 3: Print out for top 30 microbial species in UC (inflamed state)

| # Feature / Sample                  | avg_RA               |
|-------------------------------------|----------------------|
| Faecalibacterium_prausnitzii        | 0.07002881538461539  |
| Bifidobacterium_longum              | 0.06382550769230769  |
| Eubacterium_rectale                 | 0.0625186            |
| Ruminococcus_gnavus                 | 0.051036042307692306 |
| Bifidobacterium_adolescentis        | 0.04688708076923077  |
| Collinsella_aerofaciens             | 0.043927023076923075 |
| Subdoligranulum_unclassified        | 0.04151455769230769  |
| Bifidobacterium_bifidum             | 0.029528111538461542 |
| Ruminococcus_bromii                 | 0.028484765384615383 |
| Ruminococcus_sp_5_1_39BFAA          | 0.02581636153846154  |
| Ruminococcus_torques                | 0.024669315384615382 |
| Bacteroides_vulgatus                | 0.02039732307692308  |
| Streptococcus_thermophilus          | 0.01998023076923077  |
| Clostridium_nexile                  | 0.019150253846153844 |
| Ruminococcus_obeum                  | 0.014726107692307696 |
| Streptococcus_salivarius            | 0.013381111538461539 |
| Lachnospiraceae_bacterium_1_1_57FAA | 0.013357850000000001 |
| Dialister_invisus                   | 0.013128284615384615 |
| Coprococcus_comes                   | 0.012977334615384615 |
| Methanobrevibacter_smithii          | 0.011397815384615385 |
| Bacteroides_uniformis               | 0.011084892307692308 |
| Bifidobacterium_animalis            | 0.010814530769230769 |
| Eubacterium_biforme                 | 0.010752661538461538 |
| Roseburia_inulinivorans             | 0.010622423076923078 |
| Parabacteroides_unclassified        | 0.010481800000000001 |
| Eubacterium_hallii                  | 0.010479542307692307 |
| Escherichia_coli                    | 0.009268503846153844 |
| Bacteroides_stercoris               | 0.00919221153846154  |
| Dorea_longicatena                   | 0.008363092307692308 |
| Eubacterium_eligens                 | 0.007780180769230769 |

### Appendix 4: Print out for 18 species uniquely enriched in UC

These 18 species were found to be more abundant in UC than in controls

| Species                             | UC (high [f-c]) | Control (low [f-c]) |
|-------------------------------------|-----------------|---------------------|
| Faecalibacterium_prausnitzii        | 0.07003         | 0.06508             |
| Bifidobacterium_longum              | 0.06383         | 0.05768             |
| Ruminococcus_gnavus                 | 0.05104         | 0.00842             |
| Collinsella_aerofaciens             | 0.04393         | 0.03056             |
| Bifidobacterium_bifidum             | 0.02953         | 0.00729             |
| Bacteroides_vulgatus                | 0.0204          | 0.01019             |
| Streptococcus_thermophilus          | 0.01998         | 0.00682             |
| Clostridium_nexile                  | 0.01915         | 0.00043             |
| Streptococcus_salivarius            | 0.01338         | 0.00654             |
| Lachnospiraceae_bacterium_1_1_57FAA | 0.01336         | 0.00444             |
| Methanobrevibacter_smithii          | 0.0114          | 0.01119             |
| Bifidobacterium_animalis            | 0.01081         | 0.00143             |
| Eubacterium_biforme                 | 0.01075         | 0.00378             |
| Roseburia_inulinivorans             | 0.01062         | 0.00513             |
| Parabacteroides_unclassified        | 0.01048         | 0.00102             |
| Escherichia_coli                    | 0.00927         | 0.0029              |
| Eubacterium_eligens                 | 0.00778         | 0.00423             |
| Bifidobacterium_pseudocatenulatum   | 0.00197         | 0.00091             |

## Appendix 5: Print out for ratios of UC microbial species RA by control

Ratio of high [f-c] UC species RA by low [f-c] control

| Species                             | Ratio    |
|-------------------------------------|----------|
| Faecalibacterium_prausnitzii        | 1.07607  |
| Bifidobacterium_longum              | 1.10651  |
| Ruminococcus_gnavus                 | 6.06297  |
| Collinsella_aerofaciens             | 1.43731  |
| Bifidobacterium_bifidum             | 4.05124  |
| Bacteroides_vulgatus                | 2.0022   |
| Streptococcus_thermophilus          | 2.93102  |
| Clostridium_nexile                  | 44.18553 |
| Streptococcus_salivarius            | 2.04512  |
| Lachnospiraceae_bacterium_1_1_57FAA | 3.01037  |
| Methanobrevibacter_smithii          | 1.01865  |
| Bifidobacterium_animalis            | 7.54344  |
| Eubacterium_biforme                 | 2.8441   |
| Roseburia_inulinivorans             | 2.07057  |
| Parabacteroides_unclassified        | 10.26613 |
| Escherichia_coli                    | 3.19254  |
| Eubacterium_eligens                 | 1.84012  |
| Bifidobacterium_pseudocatenulatum   | 2.17005  |